

Lab Title:

Mobile Application Testing Lab (Static & Dynamic Analysis)

Objectives:

- Perform static analysis using MobSF
 - Conduct dynamic testing using Frida
 - Analyze IPC vulnerabilities using Drozer
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Tools Used:

- MobSF
 - Frida
 - Drozer
 - Android Emulator / Device
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Lab Environment:

- Target App: test.apk
 - Platform: Android
 - Testing Type: Static + Dynamic
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Test 1: Static Analysis (MobSF)

Description:

MobSF was used to decompile and analyze the APK for insecure storage and hardcoded sensitive data.

Steps Performed:

1. Uploaded APK into MobSF
2. Performed static scan
3. Reviewed storage, permissions, and code findings

Result:

Test ID	Vulnerability	Severity	Target App
016	Insecure Storage	High	test.apk

Findings:

- Sensitive data stored in plaintext (SharedPreferences / files)
 - No encryption applied
 - Possible exposure of user credentials or tokens
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Test 2: Dynamic Testing (Frida)**Description:**

Frida was used to hook application functions at runtime and bypass authentication logic.

Steps Performed:

1. Connected device/emulator
 2. Injected Frida script
 3. Hooked login/authentication function
 4. Modified return values to bypass login
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Findings:

- Authentication bypass successful
 - Weak client-side validation detected
 - Critical logic exposed at runtime
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Test 3: IPC Testing (Drozer)**Description:**

Drozer was used to analyze exported components and IPC vulnerabilities.

Findings:

- Exported activities/services accessible
- Potential unauthorized access via intents
- Lack of permission enforcement

Security Issues Identified:

- Insecure data storage
- Lack of encryption
- Client-side authentication weakness
- Improper IPC exposure

Recommendations:

Data Security:

- Encrypt sensitive data using Android Keystore
- Avoid storing credentials locally

App Security:

- Implement server-side authentication
- Use code obfuscation (ProGuard)

IPC Protection:

- Restrict exported components
 - Enforce permissions
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Conclusion:

The lab successfully demonstrated how mobile applications can be compromised through insecure storage, weak authentication, and exposed IPC components. Combining static and dynamic analysis provides a comprehensive understanding of application weaknesses and helps in building secure mobile applications. Frida was used to dynamically analyse the Android application by hooking runtime functions and modifying their behavior.

Authentication mechanisms were bypassed by altering function return values, demonstrating weak client-side validation. This highlights how attackers can manipulate app logic in real time, emphasizing the need for secure coding and server-side controls.